

Prompt Engineering on SWAYAM — Week 1 Study Material

Foundations, the 1-Page Prompt Card, Prompt Pack, Anti-Patterns, Role Switching, Mini-Project, and Interface Essentials

How to use this study material

This document is your ready reference for Week 1. It consolidates the key ideas demonstrated in Videos 1 to 8, adds structured examples (India-first, student-centric), and gives you practice tasks that build skill through repetition and reflection.

Recommended approach: read one section, run at least one prompt live, then complete the self-study exercises. The goal is not to memorise prompts; the goal is to learn how to design prompts for any situation.

Week 1 Skill Progression Map



Week 1 learning outcomes

1. Explain what Prompt Engineering is and why 'vague ask = vague output'.
2. Use the 1-Page Prompt Card to design structured prompts (Role, Audience, Context, Format, Constraints, Language).
3. Switch audience and role deliberately (academic, administrative, industry) without tone drift.
4. Apply prompt chaining to split complex tasks into clean steps.
5. Identify and fix common prompting anti-patterns using a debug checklist.
6. Build a personal Prompt Card Pack (Academic, Career, Creative) that you can reuse.
7. Complete the Week 1 mini-project (self-evaluated) and understand the weekly grading structure.

Key definitions (Week 1)

Prompt Engineering: The discipline of designing instructions so that an AI model produces outputs that are accurate, usable, and aligned to your intent.

Vague Prompt: A prompt that lacks role, audience, context, structure, or constraints—forcing the model to guess.

Prompt Card: A one-page checklist that captures the minimum information an AI needs to respond with clarity.

Prompt Chaining: Breaking a complex task into smaller steps, using each output as input for the next prompt.

Anti-Pattern: A common prompting mistake that reliably produces weak outputs (e.g., double tasks, missing constraints, tone drift).

1-Page Prompt Card (Week 1 Core Tool)

Role Who should the AI act as?	Audience Who will use the output?	Context What is the situation/background?
Format How should it be structured?	Constraints Limits, rules, do/don'ts	Language Language + tone/register

Video 1: Welcome + What Prompt Engineering is (with the 1-Page Prompt Card)

Week 1 begins with a simple realisation: if you do not specify what you need, the AI guesses—and guesses rarely match your academic or career standards. This video demonstrates the difference between vague prompts and structured prompts, then introduces the 1-Page Prompt Card as your first core tool.

The student reality: common vague prompts

Examples of what many students type (and why the output disappoints):

Write an essay on British Rule in India.
Make MCQs on GST.
LinkedIn post for my internship.

These prompts are not 'wrong'. They are incomplete. They do not tell the AI: who you are, who the output is for, what depth you need, and what structure you expect.

The core idea: Vague Prompt = Vague Output

- If you do not define the goal, the model chooses a generic goal.
- If you do not define the audience, the model chooses a default tone (often blog-style).
- If you do not define structure, the model produces long paragraphs that are hard to use.
- If you do not define constraints, you spend more time editing than learning.

The 1-Page Prompt Card (your Week 1 tool)

The Prompt Card is a checklist of six boxes. It forces clarity before you type. When you fill it, the model stops guessing and starts delivering.

```
Prompt Card (quick fill)
Role:
Audience:
Context:
Format:
Constraints:
Language:
```

Worked example: turning a vague prompt into a usable output

Vague prompt:

```
Write an essay on British Rule in India.
```

Structured prompt using the Prompt Card:

```
Role: Act as a History Teacher.
Audience: First-year BA students.
Context: How the British entered India, their key strategies, and how the
freedom movement responded.
Format: 300 words in 3 paragraphs (intro, body, conclusion).
Constraints: Simple language, plagiarism-free, no heavy jargon.
Language: Academic English.
```

Notice what changes: the output becomes shorter, structured, and aligned to your submission expectation. You can now learn from it, adapt it, and write your final answer in your own voice.

Self-study exercises (Video 1)

Complete these tasks before you move to Video 2:

- Pick one academic task you must do this week (assignment, notes, MCQs, lab record). Write a vague prompt and run it.
- Rewrite the same prompt using the 1-Page Prompt Card. Run it again.
- Compare outputs. Identify which Prompt Card field created the biggest improvement (Role, Audience, Context, Format, Constraints, Language).
- Write a one-paragraph reflection: what did you learn about your own clarity?

Video 2: Decode the Prompt Card: Role, Audience, Context, Format, Constraints, Language

This video breaks down each field of the Prompt Card 'line by line' and shows why missing even one field reduces output quality. Treat these fields as coordinates on a map: when they are complete, the model knows exactly where to go.

Role: The hat the AI should wear. A tutor, a placement mentor, a question setter, a social media manager. Role defines perspective and depth.

Audience: Who will read/use the output. A class-7 student, a first-year UG student, a postgraduate learner, a parent, a recruiter. Audience decides tone and complexity.

Context: The background: topic, scope, and situation. Context anchors relevance. Without it, the model writes generic text.

Format: The structure you want: 3 paragraphs, bullet points, 2-D table, MCQ set with answer key, poster caption, email, etc. Format reduces editing time.

Constraints: Rules and limits: word count, no emojis, avoid jargon, include 1 figure/quote, use Indian examples, cite sources. Constraints create discipline.

Language: Not only English/Hindi/Kannada, but also register: formal academic, simple English, professional tone, Hinglish, persuasive, neutral.

Quick examples across disciplines

Use cases that students across domains can relate to:

BA / Humanities

Role: Tutor | Audience: 1st-year UG | Task: Explain 'British Rule in India' as 3 key causes + 3 impacts.

BCom / Commerce

Role: Exam Question Setter | Audience: 3rd-year BCom | Task: 10 MCQs on GST basics with answer key.

BCA / BTech

Role: Placement Mentor | Audience: IT Recruiter | Task: 2 resume bullets for a React + Firebase project (quantified).

BSc / Life Sciences

Role: Lab Instructor | Audience: 1st-year BSc | Task: Photosynthesis summary in 5 bullets for lab manual.

BBA

Role: Social Media Curator | Audience: Students 18-22 | Task: Poster caption + 3 hashtags for a campus drive.

Integrated activity: fill the entire Prompt Card once

Scenario: Independence Day blood donation drive (poster caption).

Role: Social Media Manager

Audience: College students

Context: Independence Day blood donation drive on campus

Format: Poster caption + 3 hashtags
Constraints: Under 25 words, no emojis, patriotic tone
Language: English

Self-study exercises (Video 2)

- Choose one topic you studied this semester. Run two prompts: one without Audience and one with Audience. Note the difference in depth.
- Choose one output type you often need (notes, MCQs, email, report). Force a format constraint (word count + structure).
- Create one Prompt Card for a real task due this week. Save it as a reusable template in your notes.
- Optional: Ask the model to critique your Prompt Card: 'Identify what is missing, and suggest improvements without changing my intent.'

Video 3: Prompting for different audiences + Prompt Chaining

In real life, the same information must be communicated differently to different people. This video demonstrates 'audience switching' (Dean vs Parent vs Recruiter), then introduces prompt chaining as a practical method to refine outputs step by step.

Audience switching: one event, three voices

Scenario: feedback after a college hackathon or cultural fest.

Vague prompt (no audience):

Write feedback for a college hackathon we organised.

Now the same facts, three different audiences:

1) Dean / Director (formal + outcomes)
Role: Student Representative
Audience: Dean
Task: Write feedback on the hackathon. Focus on outcomes, participation data, and 3 improvement suggestions. Use formal, solution-oriented tone.

2) Parent (personal + learning)
Role: Student sharing experience
Audience: Parent
Task: Describe the hackathon experience. Highlight learning, teamwork, and what you felt proud about. Use a warm, reflective tone.

3) Recruiter (skills + impact)
Role: Student updating recruiter
Audience: Recruiter / LinkedIn professional community
Task: Write a concise LinkedIn post. Highlight skills, your contribution, and results/metrics. Professional tone.

This is not 'changing the truth'. It is changing the framing: outcomes for leadership, learning for parents, impact for recruiters.

Prompt chaining: turn one output into a workflow

When a task is multi-layered, do not force the model to do everything at once. Instead, chain prompts like steps in a report.

Example chain: Guest lecture thank-you note -> polished LinkedIn post
Prompt 1: Draft a 120-word thank-you email for a guest lecture (formal, neutral).
Prompt 2: Add 3 specific details: speaker bio, topic highlights, and one student takeaway.
Prompt 3: Convert the refined email into an 80-word LinkedIn post with 3 hashtags (no emojis).

Self-study exercises (Video 3)

- Pick one real event you attended this month (seminar, fest, guest lecture, sports event). Create three outputs: Dean, Parent, Recruiter. Keep facts same; change framing.
- Do one prompt chain of at least 3 steps for any task you care about (notes -> summary -> WhatsApp message; or email -> LinkedIn post -> poster caption).
- Write a small rule for yourself: 'When my prompt has more than one task, I will split and chain.'

Prompt Card template (printable quick reference)

Field	What it controls	Ask yourself (student lens)	Example fill (compact)
Role	Perspective + expertise	Who should AI act as?	Tutor / Question Setter / Placement Mentor
Audience	Tone + depth	Who will read this?	1st-year UG / Parent / Recruiter
Context	Relevance	What is my exact situation?	Topic + syllabus + my deliverable
Format	Structure	In what output form?	3 paras / 5 bullets / 10 MCQs + key
Constraints	Discipline	What are my rules?	≤300 words; no emojis; simple English
Language	Connection	How should it sound?	Academic / professional / Hinglish

Video 4: Hands-on: Build your Prompt Card Pack

This video upgrades your mindset: prompt engineering is not only about getting outputs. It is about learning to think structurally. You build three reusable Prompt Cards—Academic, Career, Creative—and you learn the deeper idea: good prompts are design artifacts.

Prompt engineering as design thinking

Use this thinking loop when designing any prompt:

- Empathise: What does the user/audience actually need?
- Define: What is the real problem statement (not the symptom)?

- Ideate: What structure and constraints will produce a usable output?
- Test: Run v1, refine to v2, and observe what changed.

Your three core cards (Week 1)

Build these three cards and store them as reusable templates:

Academic Card (analysis-first)
 Role: Subject Tutor / Scholar
 Audience: Your class level (UG/PG)
 Context: Topic + scope + what you want to understand
 Format: Notes / 3-paragraph essay / 2-D table / revision sheet
 Constraints: word limit; exam-focused; examples from India where possible
 Language: Academic but simple

Career Card (positioning-first)
 Role: Career Strategist / Recruiter lens
 Audience: Recruiter / hiring manager
 Context: project/internship with measurable outcomes
 Format: resume bullets + short LinkedIn post
 Constraints: quantified impact; avoid buzzwords; no emojis
 Language: Professional English

Creative Card (emotion-first)
 Role: Social Media Curator / Campaign planner
 Audience: Students (18-22) or public
 Context: fest/campaign with theme and call-to-action
 Format: caption + hashtags / 30-sec script / poster text
 Constraints: very short; high clarity; positive tone
 Language: crisp, energetic (but clean)

Meta-learning: three ways of thinking

Academic prompts train causal thinking (why did it happen?), career prompts train value thinking (why does it matter?), and creative prompts train emotional thinking (why should someone care?). This is why the Prompt Pack is more than templates—it is cognitive training.

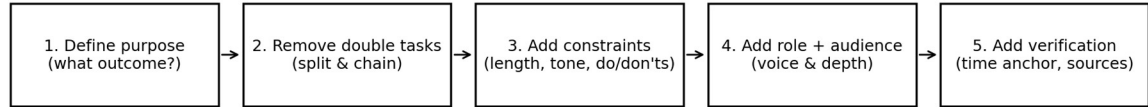
Self-study exercises (Video 4)

- Create one Academic Card for a topic you struggle with. Run it twice: v1 and v2 (tighten constraints in v2).
- Create one Career Card for a project you have built. Produce 2 resume bullets and one LinkedIn post.
- Create one Creative Card for a campus event you care about. Produce a poster caption + 3 hashtags.
- Save all three cards in a single document titled: 'My Prompt Pack - Week 1'.

Video 5: Common Pitfalls & Anti-Patterns in Prompt Design

Strong prompting is not only about writing good prompts; it is also about debugging bad ones. This video introduces five anti-patterns that commonly fail in student life, and shows how to fix them with small, disciplined edits.

Prompt Doctor: 5-Step Debug Checklist (Video 5)



The five anti-patterns (with fixes)

Anti-pattern 1: Vague prompt —

- Symptom: generic answer. Fix: add role + audience + context + constraints.

Anti-pattern 2: Double task —

- Symptom: half-summary, half-analysis. Fix: split tasks and chain prompts.

Anti-pattern 3: Missing constraints —

- Symptom: tone chaos or unusable length. Fix: word limit + structure + do/don'ts.

Anti-pattern 4: Tone drift —

- Symptom: casual voice when you need academic voice. Fix: tone guardrails (avoid casual phrases/exclamations).

Anti-pattern 5: Fact drift —

- Symptom: outdated or uncertain data. Fix: time anchor + ask to flag source year + verify.

Prompt Doctor checklist (apply to any weak output)

- Purpose: What exact outcome do I need (notes, MCQs, email, plan, etc.)?
- Single-task: Is my prompt trying to do two tasks? If yes, split and chain.
- Constraints: Did I specify length, tone, and do/don'ts?
- Role + audience: Did I define the voice and the reader?
- Verification: Did I anchor time (year/semester) and ask for source clarity for facts?

Self-study exercises (Video 5)

- Find one prompt you used earlier that disappointed you. Diagnose it using the five anti-patterns. Rewrite and rerun.
- Take a double-task prompt from your domain (e.g., 'summarise and analyse'). Split it into a 2-step chain.
- Write one sentence rule: 'Constraints do not restrict creativity; they focus it.' Apply it to one creative prompt.

Video 6: Live prompting demos: Academic vs Administrator vs Industry roles

This video shows how a single AI model changes its voice when you define roles. You learn to create outputs that match the room you are in: classroom (explanatory), office (procedural), boardroom/LinkedIn (impact-driven).

Role-to-voice mapping (remember this table)

Role	Voice	Focus	Typical student use
Academic	Explanatory, evidence-based	Clarity of understanding	Essays, reports, exam notes
Administrator	Neutral, procedural	Compliance and clarity	Notices, emails, circulars
Industry	Crisp, impact-driven	Results and reflection	Portfolios, LinkedIn, pitches

Worked prompts (one question, three roles)

Topic example: 'AI in climate research in India' (BSc report).

Academic Role Prompt
Role: Environmental Science Tutor
Audience: Undergraduate students
Context: How AI aids climate change research in India (IMD, ISRO, etc.)
Format: 3 sections (Problem, AI applications, Indian case example)
Constraints: 600 words; cite recent reports (2020 onwards)
Language: Academic English

Administrative example: project submission deadline notice.

Administrator Role Prompt
Role: Academic Administrator
Audience: UG and PG students
Context: Semester-end project report submission
Format: Formal notice (<=120 words) with subject line + signature line
Constraints: Include deadline and procedure; no emojis; neutral tone
Language: Professional English

Industry example: internship completion post.

Industry Role Prompt
Role: Marketing Professional
Audience: LinkedIn peers, recruiters, mentors
Context: Internship completed; quantify impact (e.g., reduced reporting time by 30%)
Format: 90-word LinkedIn post + 2 hashtags
Constraints: Start with achievement metric; avoid overused words like 'great'
Language: Professional English

Self-study exercises (Video 6)

- Pick one topic from your life (a project, event, or concept). Write three prompts: Academic, Administrator, Industry. Run and compare.

- Identify what changed the most: vocabulary, structure, or focus (knowledge vs compliance vs impact).
- Rewrite one prompt to be more responsible: remove ambiguity, include deadline, include measurable outcomes.

Video 7: Mini-project brief & self-evaluation parameters (Week 1 deliverable)

This is your first 'prompt prototype'. The mini-project moves you from knowing concepts to building something measurable. You pick one real problem, write three prompt versions (vague -> structured -> role-based), and evaluate your improvement.

Choose one track

- Academic Track: solve a learning or research challenge (notes, abstract, paper summary, exam prep).
- Administrative Track: simplify a campus/dept process or communication (notice, email, circular, form guidance).
- Career Track: create a professional artefact (resume bullets, LinkedIn post, portfolio intro, interview answer).

Your mini-project steps (Version 1 to Version 3)

- Define the problem in one sentence (your real context).
- Version 1 (Vague): write the first prompt that comes to mind; save the output.
- Version 2 (Structured): apply Prompt Card (format + constraints + context); save output.
- Version 3 (Role-based): customise role and audience sharply; add measurable requirements; save output.
- Write a short reflection: what improved and why?

Self-evaluation parameters (5-point scale: 0–1, 2–3, 4–5)

Use these parameters to evaluate each prompt version and see your progress:

Parameter	0–1	2–3	4–5
Clarity of intent	0–1: unclear task; AI guesses	2–3: mostly clear but some ambiguity	4–5: crystal-clear purpose and deliverable
Completeness (Prompt Card coverage)	0–1: missing 3+ fields	2–3: covers 4–5 fields	4–5: covers all 6 fields meaningfully
Structure and usability of output	0–1: long/generic, hard to use	2–3: somewhat structured, needs editing	4–5: ready-to-use, minimal edits
Role + audience alignment	0–1: wrong tone/level	2–3: partially aligned	4–5: perfectly aligned voice and depth
Verification and responsibility	0–1: no constraints/accuracy checks	2–3: some constraints but weak checks	4–5: strong constraints, time anchors, asks for source clarity

Sample mini-project (short)

Career track example: convert a project into recruiter-ready bullets.

V1 (vague): Write resume points for my project.

V2 (structured): Role: Placement mentor | Audience: recruiter | Context: React + Firebase attendance app | Format: 2 bullets | Constraints: quantify impact, action verbs, <=25 words each.

V3 (role-based): Role: Tech recruiter | Audience: campus hiring panel | Context: built attendance app used by 60 students; reduced manual attendance effort by 40% | Format: 2 bullets + 1-line tech stack | Constraints: ATS-friendly, no buzzwords.

Your goal is not to copy this. Your goal is to build your own prompt evolution for your own context.

Video 8: Recap, assignment walkthrough, and interface essentials

This closing video does two things: (1) it recaps the learning journey of Week 1, and (2) it highlights that tool literacy matters—the interface features (personalisation, connectors, formatting) influence output quality.

Weekly assessment structure

Each week includes a graded self-assignment (25 percent) and the remaining 75 percent comes through the end-term proctored exam on SWAYAM. Your Week 1 assignment is built around your mini-project: refine it, apply the rubric, and submit as per portal instructions.

Interface best practices for better responses

- Start fresh for a new topic to avoid context bleed-over.
- Use numbered lists and bullets for output format requirements.
- Embed one example of the kind of output you want (models learn patterns).
- Ask the model to flag uncertain facts and to mention the year/source where possible.
- When you need factual accuracy, cross-check with reliable sources.

Personalisation and connectors (conceptual overview)

Personalisation helps the tool adapt to your preferences (tone, level, repeated needs). Connectors (where available) allow the model to work with your files and apps—useful for summarising notes, drafting reports, or extracting structure from documents. Use these responsibly: do not share private data unnecessarily, and always verify outputs that carry academic or legal risk.

Self-study exercises (Video 8)

- Open your chat interface settings and note what personalisation options exist.
- Practice one prompt where you explicitly define tone and length (e.g., 'formal academic English, 150 words').
- Take your mini-project Version 3 prompt and run one final refinement step: 'Improve clarity without changing meaning; keep constraints intact.'
- Write a 5-line recap of what Week 1 changed in your prompting behaviour.

Week 1 quick reference (one-page recap)

- If output is weak: run Prompt Doctor (purpose -> single-task -> constraints -> role/audience -> verify).
- Always specify: Role + Audience + Context + Format + Constraints + Language.
- If prompt has two tasks: split and chain.
- Build a Prompt Pack: Academic, Career, Creative.
- Measure progress through mini-project versions (V1 -> V2 -> V3) and reflect using the 5-point parameters.

Appendix A: Reusable prompt templates

Template 1: Notes for exams (any subject)

Role: Subject tutor
Audience: My semester level (UG/PG)
Context: Topic + syllabus unit + what I struggle with
Format: 10 bullet points + 5 key terms + 2 examples
Constraints: <=300 words; simple language; include 1 mnemonic
Language: Academic but simple

Template 2: MCQs with answer key

Role: Exam question setter
Audience: Students of my programme/semester
Context: Topic + difficulty level (easy/medium/mixed)
Format: 10 MCQs, 4 options each, answer key at end
Constraints: Avoid ambiguous options; no trick questions unless asked
Language: Clear English

Template 3: Convert content to multiple outputs (prompt chain)

Step 1: Summarise the document in 5 bullets (<=150 words).
Step 2: Extract 5 action points with deadlines (if any).
Step 3: Convert into a WhatsApp message for classmates (<=80 words).
Step 4: Convert into a formal email to faculty (<=120 words, no emojis).